

Scalability & Future Plan

Date	1 October 2026
Team ID	SWTID-2026-8336
Project Name	EduGenie — Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Dependence on Gemini API availability and usage limits	AI responses may be affected if the API is unavailable or usage limits are reached	High
2	AI-generated responses may occasionally require verification	Students may receive inaccurate or incomplete information for some questions	High
3	Current system is designed primarily for a limited-scale student user base	Performance may need optimization when the number of concurrent users increases	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed for a small to moderate number of users	Deploy scalable backend services and use load balancing to support more concurrent users
2	Data Storage	MongoDB is used to store user and learning-related data	Increase database capacity and use indexing, optimizer queries, and suitable MongoDB scaling options
3	Performance	REST API and Gemini requests are processed through the backend	Implement caching, asynchronous processing, API optimization, and monitoring
4	Security	JWT-based authentication and protected API endpoints	Add stronger validation, HTTPS, rate limiting, secure token handling, and improved access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
-------	-------------------------------	-----------------	-----------------

Phase 1	Improve AI tutor, topic explanations, and quiz generation	1-2 months	Better personalized learning experience
Phase 2	Add personalized learning paths and student progress analytics	3-4 months	Help students track their learning and identify areas for improvement
Phase 3	Add multilingual learning support and voice-based interaction	5-6 months	Improve accessibility and support more learners
Phase 4	Cloud scaling, advanced analytics, and intelligent recommendation features	6-12 months	Support larger user volumes and provide more personalized learning